

# Project Problem Statement

Problem Statement No. 22 – Interview Trainer Agent

Candidate: Sachin Kumar

## 1. The Challenge

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Job seekers often struggle to prepare effectively for competitive interviews due to generic question banks, lack of personalized feedback, and limited understanding of specific company and role expectations. Real-time assessment of both technical proficiency and soft skills is difficult to access without expensive professional coaching. Without dynamic, role-tailored simulation and actionable evaluation, candidates struggle to identify skill gaps and build interview confidence.

## 2. The Objective

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Build an AI-powered **Interview Trainer Agent** using Retrieval-Augmented Generation (RAG) and IBM Granite to deliver realistic, personalized mock interviews and preparation strategies. The system will retrieve role-specific interview questions, industry expectations, behavioral scenarios, and HR guidelines from recruitment portals, professional networks, and company databases.

## 3. Key System Features & Capabilities

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- **Tailored Question Generation:** Users can input their resume or job title, and the agent provides targeted questions based on identified skill gaps and experience levels.
- **RAG Knowledge Retrieval:** Fetches actual industry expectations, HR guidelines, and behavioral scenarios from external databases to ground the AI responses.
- **Comprehensive Assessment:** Supports both technical (coding/domain knowledge) and soft skill (behavioral/STAR method) assessments for a complete interview experience.
- **Multi-Agent Architecture:**
  - **Profile Agent:** Analyzes resumes, experience levels, and target job roles.
  - **Question Agent:** Retrieves and adapts role-specific questions.
  - **Evaluation & Feedback Agent:** Analyzes answers, provides scores (out of 10), highlights missing points, and generates ideal model answers.
- **Advanced Modalities:** Integrates voice-based answering (speech-to-text) and anti-cheat mechanisms (disabling copy-paste, focus tracking) for a realistic interview environment.
- **Dashboard & Roadmap:** Provides visual analytics and generates a downloadable personalized preparation roadmap based on interview performance.

## 4. Mandatory Technology Stack

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The solution must strictly utilize the following infrastructure:

- **IBM Cloud Lite services:** For hosting, scaling, and orchestrating workflows.
- **IBM Granite Model:** To power the core AI generation, reasoning, model answers, and natural language understanding capabilities.
- **IBM watsonx.ai:** For model deployment, embeddings, and secure AI governance.